

Symmetric Functions and Review on Exponents and Logarithms (Homework)

Each table below has points collected from a graph of an even function. Fill the empty spaces with other points from the graph.

①

x	7	4	2	3	15				
f(x)	5	1	10	7	4				

②

x	0	-1	$\frac{2}{3}$	0.1	-5				
f(x)	3	4	0	-7	-1				

③ If "f" is an even function, and $f(2)=10$ find $f(-2)$.

④ If "f" is an even function, and $f(-17)=0$, find $f(17)$.

Each table below has points collected from a graph of an odd function. Fill the empty spaces with other points from the graph.

⑤

x	-3	36	-2	12	0				
f(x)	8	-1	-2	10	0				

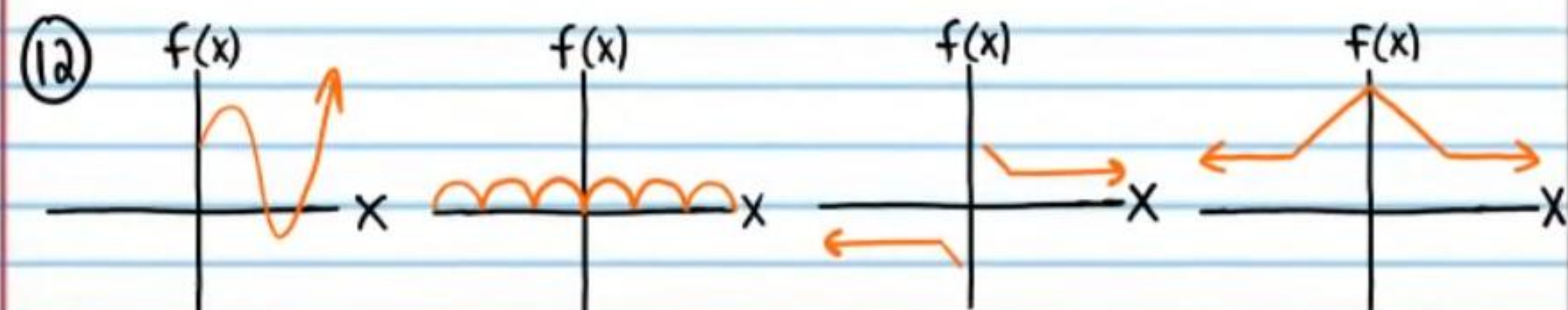
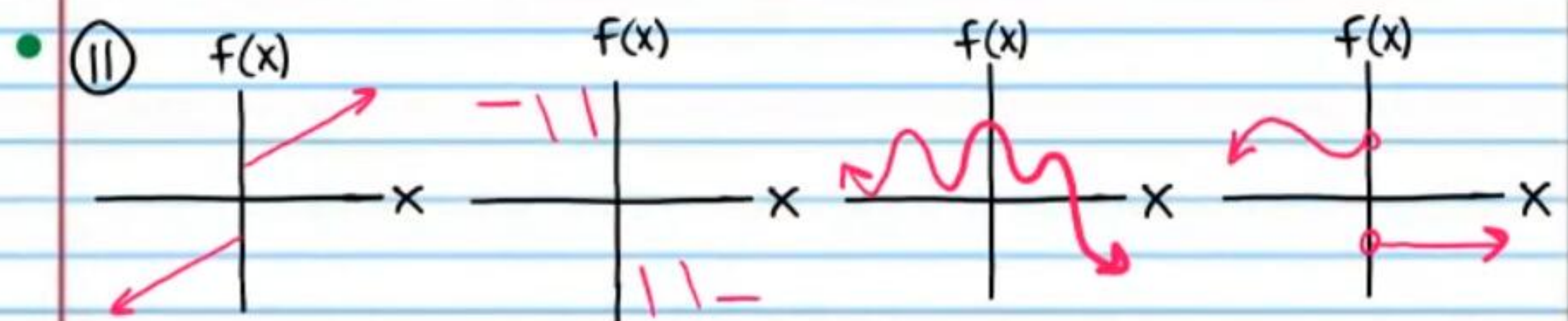
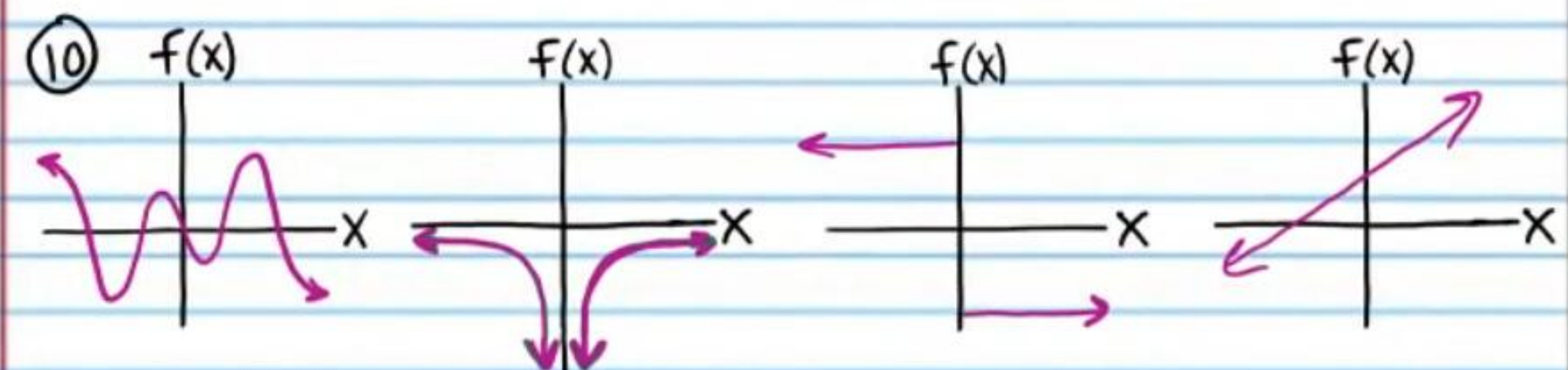
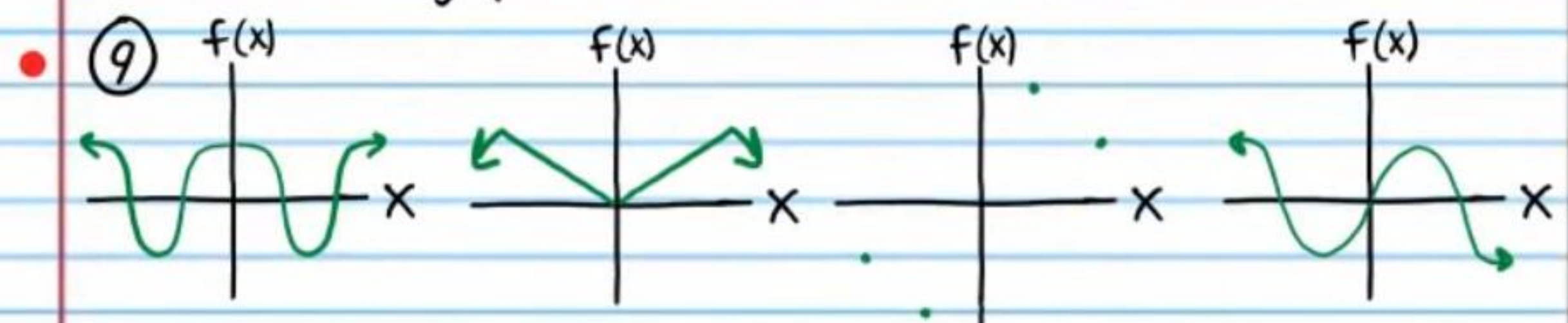
⑥

x	19	67	42	-11	13				
f(x)	14	30	-9	18	-7				

⑦ If "f" is an odd function, and $f(-4.7)=-12$, find $f(4.7)$.

⑧ If "f" is an odd function, and $f(42)=-100$, find $f(-42)$.

Determine if the graphs below are even functions, odd functions, or neither.



$$\sin(-\theta) = -\sin\theta \quad \cos(-\theta) = \cos\theta \quad \tan(-\theta) = -\tan\theta$$

$$\csc(-\theta) = -\csc\theta \quad \sec(-\theta) = \sec\theta \quad \cot(-\theta) = -\cot\theta$$

Show that each function below is an even function.

$$(13) f(x) = -10x^2$$

$$(14) f(x) = |\csc x|$$

$$(15) f(x) = -\cos x - x^6$$

$$(16) f(x) = \frac{\cot x + 10x}{\sin x + 3x}$$

Show that each function below is an odd function.

$$(17) f(x) = \sin x$$

$$(18) f(x) = \frac{|x|}{x^3}$$

$$(19) f(x) = x - x^5 - \frac{1}{2}x^7 + \tan x \quad (20) f(x) = \cot + \sqrt[3]{x}$$

Show that each function below is neither odd nor even.

②① $f(x) = \cos x - \csc x$

If even, then

$$f(-x) = f(x)$$

$$f(-x) = \cos x - \csc x$$

If odd, then

$$f(-x) = -f(x)$$

$$f(-x) = -(\cos x - \csc x)$$

$$f(-x) = -\cos x + \csc x$$

Now let's check:

②② $f(x) = 8x^9 + 3x^7 + 4$

If even, then

$$f(-x) = f(x)$$

$$f(-x) = 8x^9 + 3x^7 + 4$$

If odd, then

$$f(-x) = -f(x)$$

$$f(-x) = -(8x^9 + 3x^7 + 4)$$

$$f(-x) = -8x^9 - 3x^7 - 4$$

Now let's check:

②③ $f(x) = 10^{-x}$

If even, then

$$f(-x) = f(x)$$

$$f(-x) = 10^{-x}$$

If odd, then

$$f(-x) = -f(x)$$

$$f(-x) = -10^{-x}$$

Now let's check:

Solve.

②④ $4^{3x} \cdot 2^{3x} = \frac{3 \cdot 2^{x-1}}{4^{x-1}}$

②⑤ $1 = \frac{\sqrt[3]{6^x}}{6^{2x}}$

②⑥ $3^{x+1} \cdot 3 = \frac{1}{9^x}$

Determine if you can solve the following equations without a calculator. If you need a calculator, write the solution in $\log_b C$ notation, and then use a calculator to approximate the answer to the nearest ten thousandth.

(27) $49 = 7^x$ (28) $3^x = 5$ (29) $6 = 3(2^x)$

(30) $\frac{5^x}{2} - 4 = 3$ (31) $16 = \frac{8^x}{3} + 1$ (32) $-6(9^x) + 7 = -479$

Solve.

(33) $7 \log_5 X - 1 = 0$

(34) $6 = 5 - \frac{\log_3 X}{2}$

(35) $9 - 3 \log_6 X = 6$

(36) $\frac{\log_8 X}{10} + \frac{1}{20} = \frac{1}{5}$

Homework Answers

①

x	7	4	2	3	15	-7	-4	-2	-3	-15
f(x)	5	1	10	7	4	5	1	10	7	4

②

x	0	-1	$\frac{2}{3}$	0.1	-5	0	1	$-\frac{2}{3}$	-0.1	5
f(x)	3	4	0	-7	-1	3	4	0	-7	-1

③ 10

④ 0

⑤

x	-3	36	-2	12	0	3	-36	2	-12	0
f(x)	8	-1	-2	10	0	-8	1	2	-10	0

⑥

x	19	67	42	-11	13	-19	-67	-42	11	-13
f(x)	14	30	-9	18	-7	-14	-30	9	-18	7

⑦ 12

⑧ 100

⑨ Even Even Odd Odd

⑩ Odd Even Odd Neither

⑪ Odd Odd Neither Neither

⑫ Neither Even Odd Even

⑬ $f(x) = -10x^2$

$$f(-x) = -10(-x)^2$$

$$f(-x) = -10x^2$$

$$f(-x) = f(x)$$

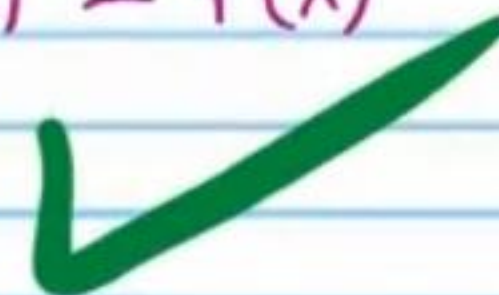


⑮ $f(x) = -\cos x - x^6$

$$f(-x) = -\cos(-x) - (-x)^6$$

$$f(-x) = -\cos x - x^6$$

$$f(-x) = f(x)$$



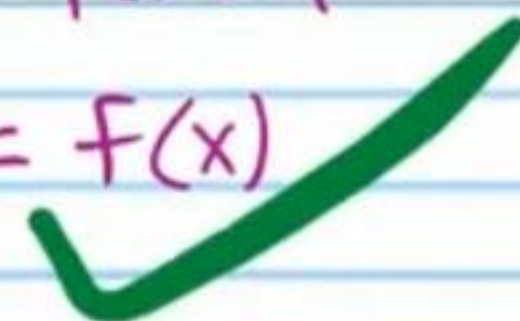
⑭ $f(x) = |\csc x|$

$$f(-x) = |\csc(-x)|$$

$$f(-x) = |-\csc x|$$

$$f(-x) = |\csc x|$$

$$f(-x) = f(x)$$



⑯ $f(x) = \frac{\cot x + 10x}{\sin x + 3x}$

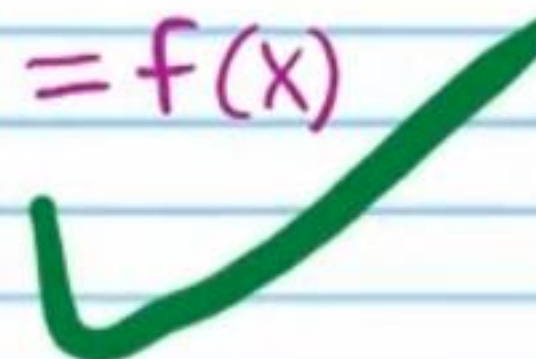
$$f(-x) = \frac{\cot(-x) + 10(-x)}{\sin(-x) + 3(-x)}$$

$$f(-x) = \frac{-\cot x - 10x}{-\sin x - 3x}$$

$$f(-x) = \frac{-(\cot x + 10x)}{-(\sin x + 3x)}$$

$$f(-x) = \frac{\cot x + 10x}{\sin x + 3x}$$

$$f(-x) = f(x)$$

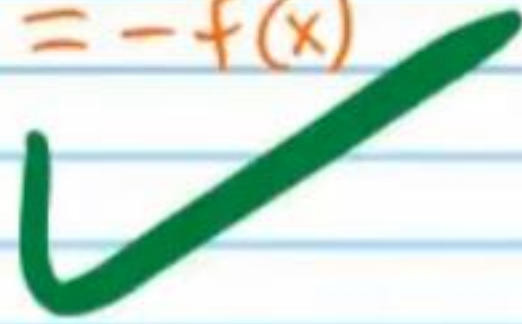


$$(17) f(x) = \sin x$$

$$f(-x) = \sin(-x)$$

$$f(-x) = -\sin x$$

$$f(-x) = -f(x)$$



$$(18) f(x) = \frac{|x|}{x^3}$$

$$f(-x) = \frac{|-x|}{(-x)^3}$$

$$f(-x) = \frac{|x|}{-x^3}$$

$$f(-x) = -\frac{|x|}{x^3}$$



$$(19) f(x) = x - x^5 - \frac{1}{2}x^7 + \tan x \quad (20) f(x) = \cot + \sqrt[3]{x}$$

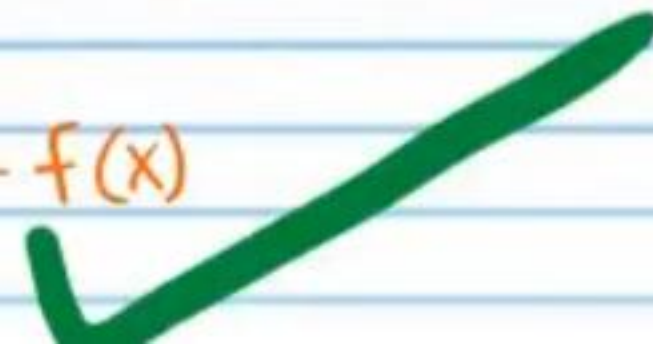
$$f(-x) = -x - (-x)^5 - \frac{1}{2}(-x)^7 + \tan(-x)$$

$$f(-x) = -x - (-x^5) - \frac{1}{2}(-x^7) - \tan x$$

$$f(-x) = -x + x^5 + \frac{1}{2}x^7 - \tan x$$

$$f(-x) = -(x - x^5 - \frac{1}{2}x^7 + \tan x)$$

$$f(-x) = -f(x)$$



$$f(-x) = \cot + \sqrt[3]{-x}$$

$$f(-x) = \cot(-\sqrt[3]{x})$$

$$f(-x) = -\cot + \sqrt[3]{x}$$

$$f(-x) = -f(x)$$



$$(21) f(x) = \cos x - \csc x \quad (22) f(x) = 8x^9 + 3x^7 + 4$$

$$f(-x) = \cos(-x) - \csc(-x) \quad f(-x) = 8(-x)^9 + 3(-x)^7 + 4$$

$$f(-x) = \cos x + \csc x$$

$$f(-x) = 8(-x^9) + 3(-x^7) + 4$$



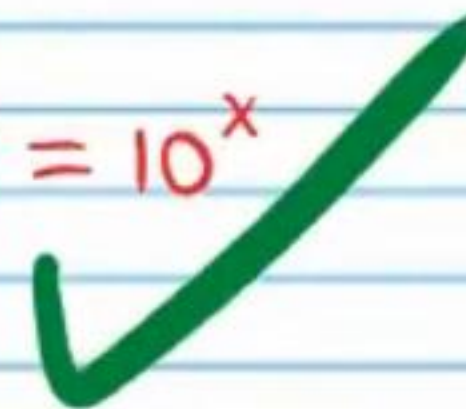
$$f(-x) = -8x^9 - 3x^7 + 4$$



$$(23) f(x) = 10^{-x}$$

$$f(-x) = 10^{-(-x)}$$

$$f(-x) = 10^x$$



$$(24) \boxed{-\frac{1}{2}}$$

$$(25) \boxed{0}$$

$$(26) \boxed{-\frac{2}{3}}$$

$$(27) \boxed{2}$$

No Calculator

$$(28) \log_3 5 \approx 1.465$$

Calculator

$$(29) \boxed{1}$$

No Calculator

$$(30) \log_5 14 \approx 1.6397$$

Calculator

$$(31) \log_8 45 \approx 1.8306$$

Calculator

$$(32) \boxed{2}$$

No Calculator

$$(33) \boxed{\sqrt{5}} \quad (34) \boxed{\frac{1}{9}}$$

$$(35) \boxed{6} \quad (36) \sqrt{8^3} \text{ or } (\sqrt{8})^3 \text{ or } 16\sqrt{2}$$